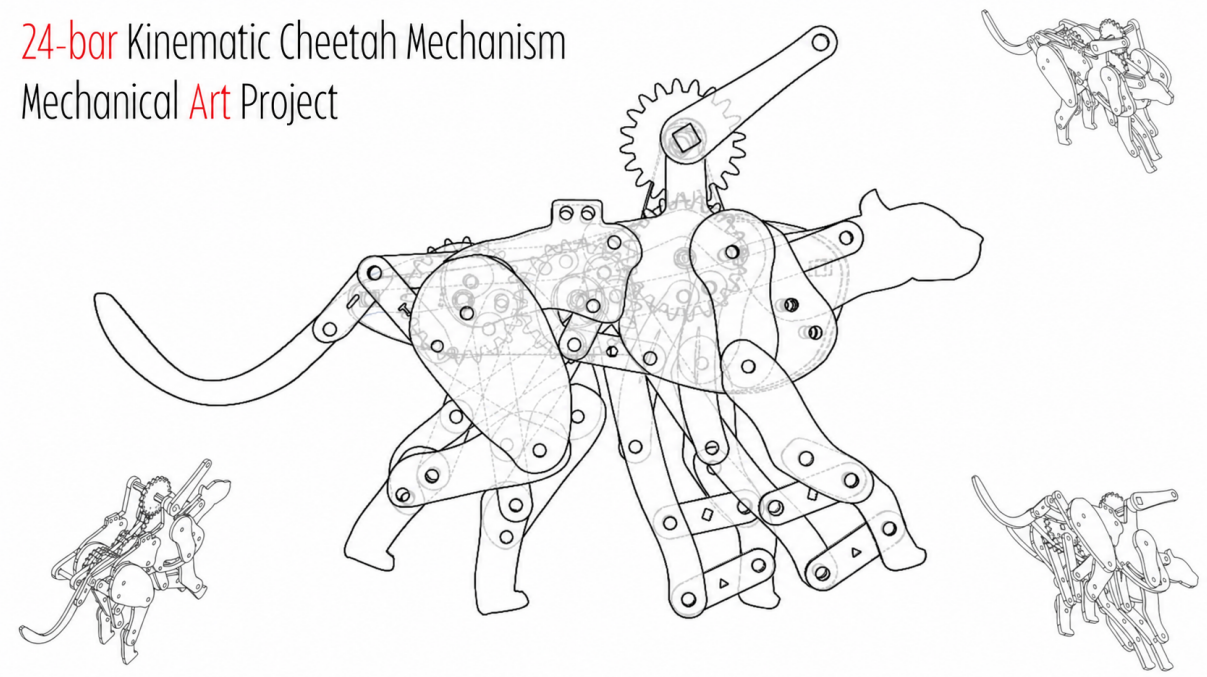
A personal project in kinetic art

24-Bar Kinematic Mechanism with 1-DOF Dual-Phase Coupling for Gait Motion Synthesis

REPORT | Shobit Verma



Personal kinetic art project: cheetah form, linkage assembly and cyclic movement.

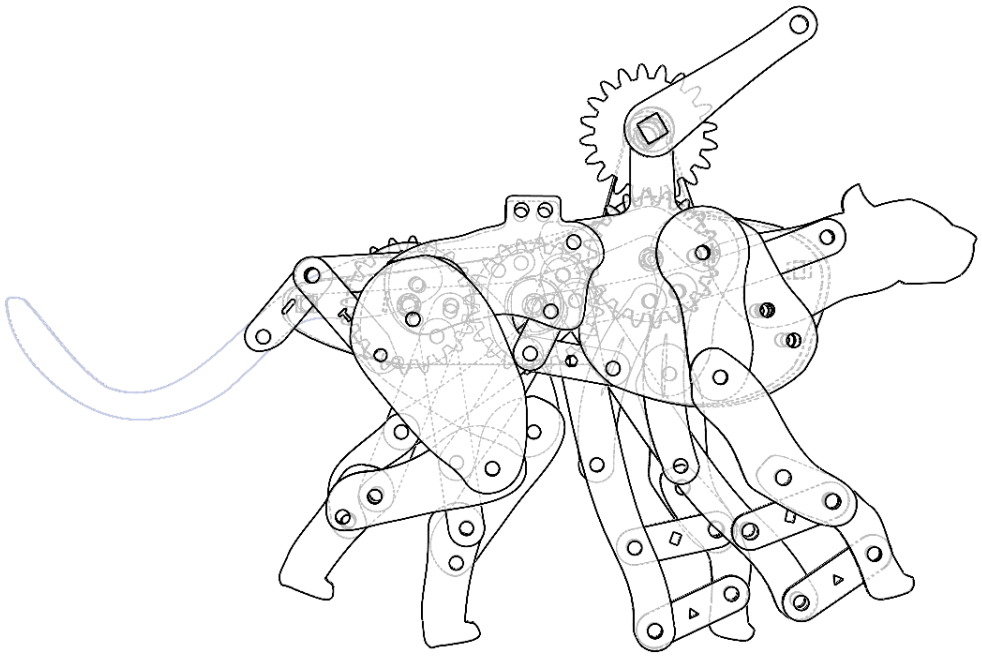
Project overview

This project explores the gallop of a cheetah as a moving mechanical sculpture. Six reference poses and their traced points guide a linkage design that turns one rotary input into coordinated leg movement. The mechanism uses a 24-bar arrangement with mirrored linkage sets, revolute joints and gear-driven phase coupling. The work includes path studies, CAD models, kinematic demonstrations and a first CNC-cut prototype.

The project combines gait observation, path synthesis, mechanical design and prototyping. Indian design application 519665-001 accompanies the report.

1. Artistic intent and design scope

The artistic aim is to make a running animal recognisable through motion as well as outline. A repeating cycle should express gathering beneath the body, reaching forward and returning through recovery. The cheetah silhouette, tail, layered plates and mechanically coordinated legs are developed as parts of the same kinetic composition.



Side view of the sculpture and mechanical linkage.

Single-input motion

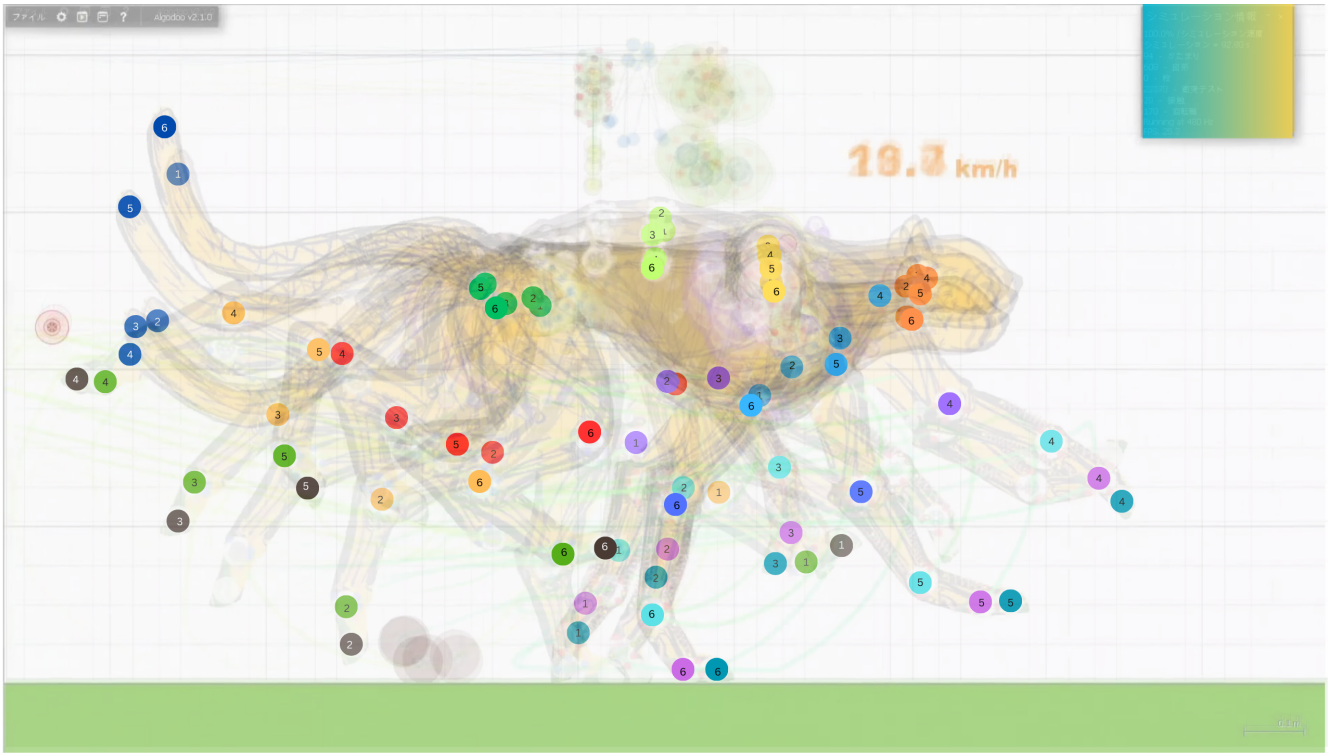
The mechanism is driven by one input. Linkages distribute that motion to the legs and gears maintain the relationship between the mirrored sides. This reduces the number of independent drives required to produce the basic repeating gait.

Design development

The project includes a gallop reference video, six pose pages, a combined path image, front and rear linkage studies, CAD views, CNC-cut component photographs and prototype working videos. These stages connect artistic intent with mechanical development.

2. Gallop reference and tracked points

The motion reference is sampled into six successive poses. The points selected on the body and limbs describe the changing geometry of the animal throughout that cycle. Their positions are image-plane references rather than measured physical coordinates.

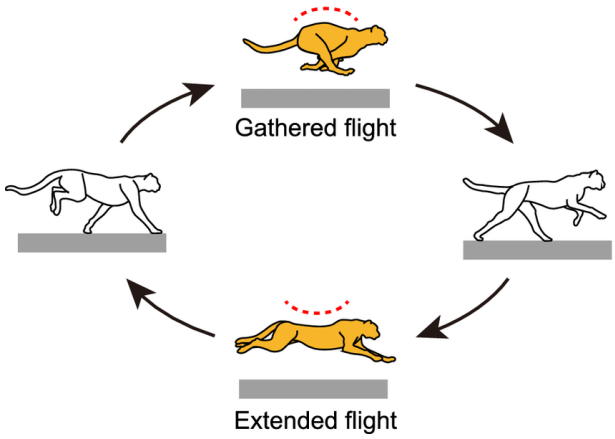


Combined gait reference. Circle colour identifies the reference track; numerals identify poses 1–6.

Colour convention

Each tracked point has a consistent colour throughout the gait sequence. Numbers identify poses 1 to 6, allowing the position of each point to be followed through the cycle.

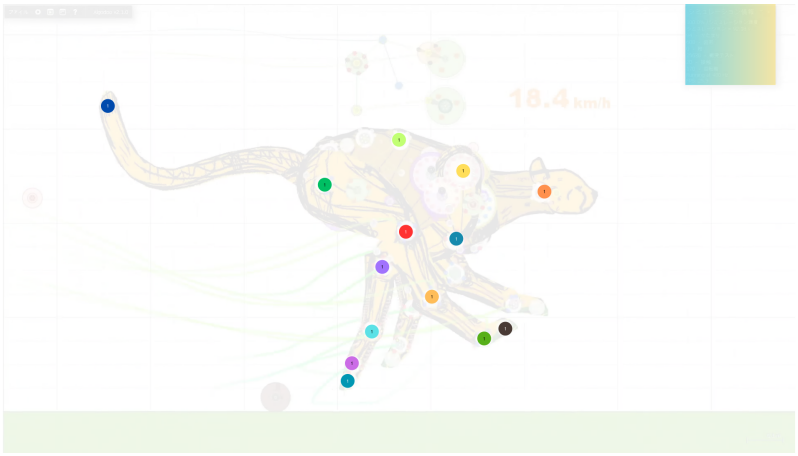
The individual poses and combined view show the same sequence of tracked positions.



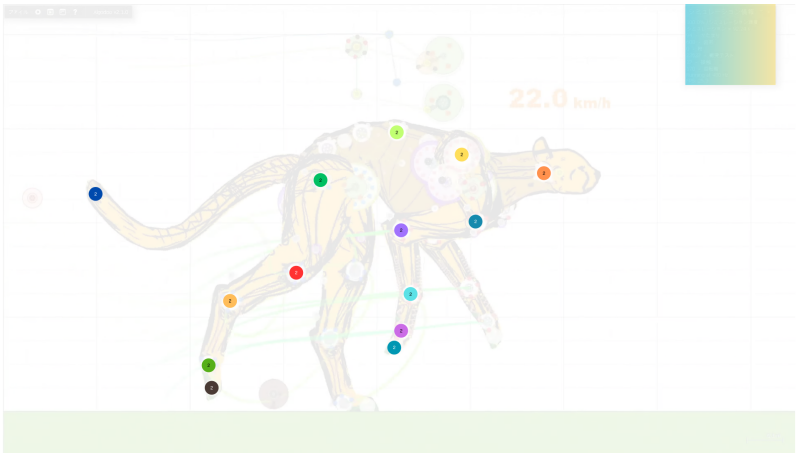
Gathered and extended flight phases of the gallop.

3. Reference poses 1–3

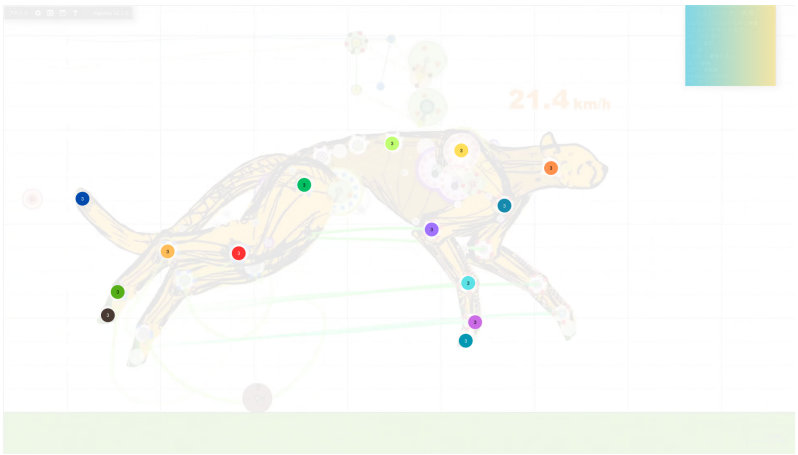
Colour follows the tracked point across the sequence. The number inside each circle identifies the pose.



Pose 1: gait reference.



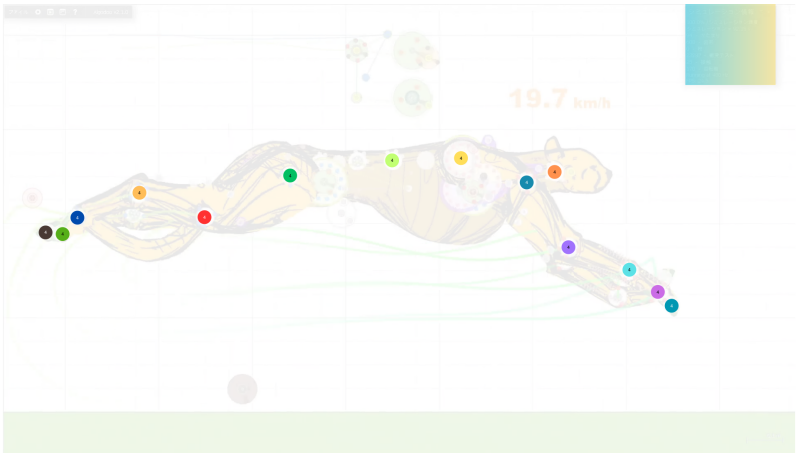
Pose 2: gait reference.



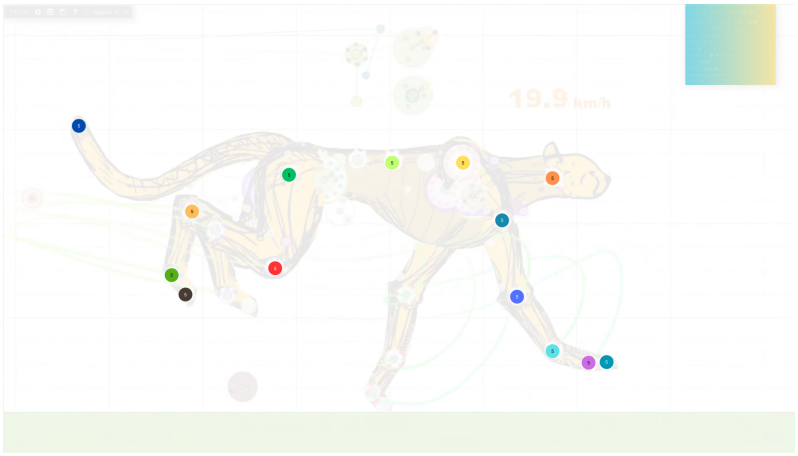
Pose 3: gait reference.

4. Reference poses 4–6

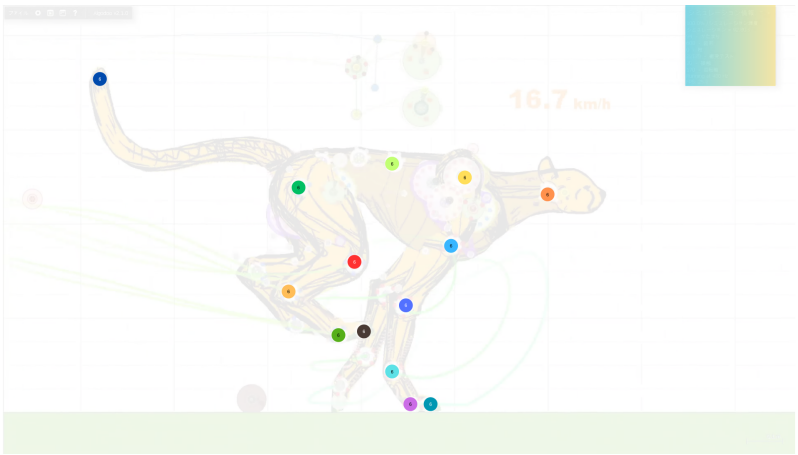
The same colour convention continues through the remaining positions of the gallop.



Pose 4: gait reference.



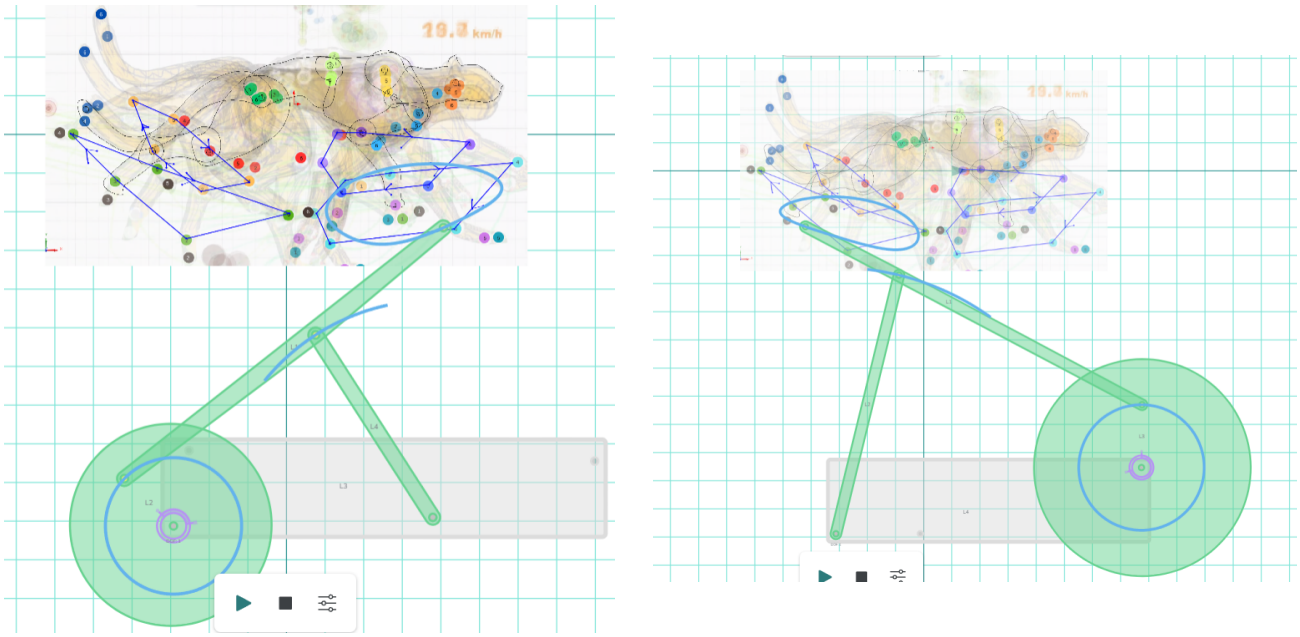
Pose 5: gait reference.



Pose 6: gait reference.

5. Front- and rear-leg path synthesis

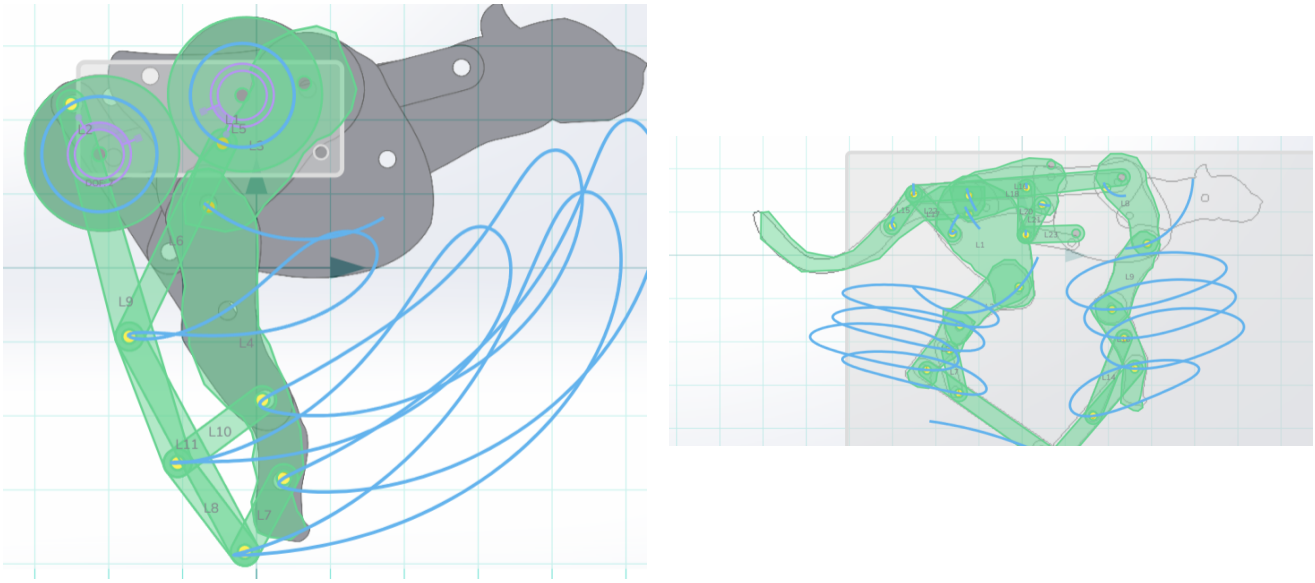
The reference paths guide the shape of the leg movement. A useful cyclic path must include forward reach, a lower part of the motion and recovery into the next step. The front- and rear-leg studies show the reference image combined with linkage geometry and traced curves.



front- and rear-leg path studies.

Front-leg refinement

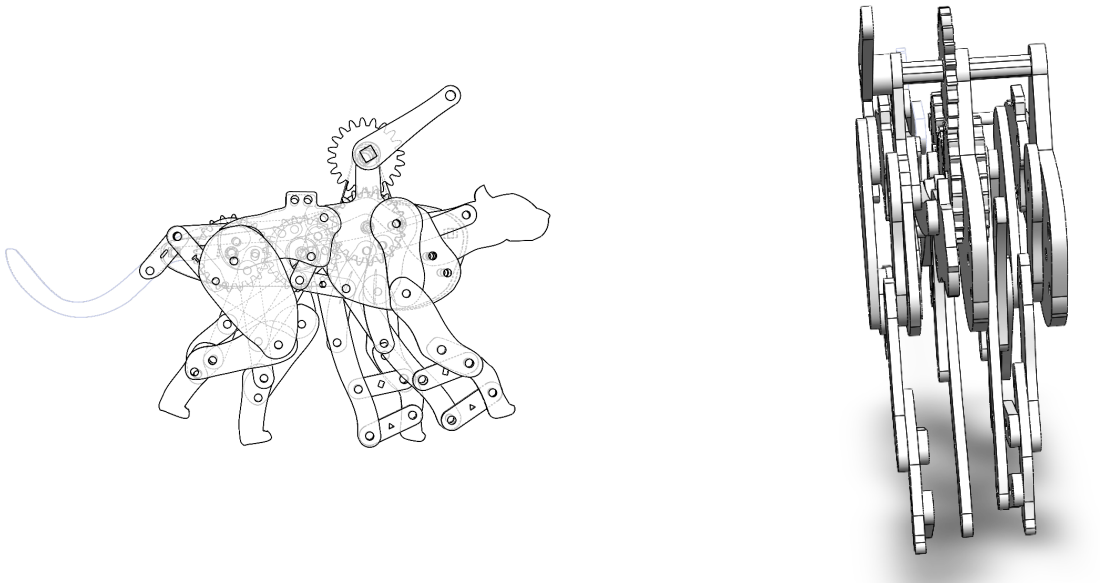
Front-leg geometry is refined by visually comparing the linkage trajectory with the reference path. Trajectory error and optimisation performance have not been quantified.



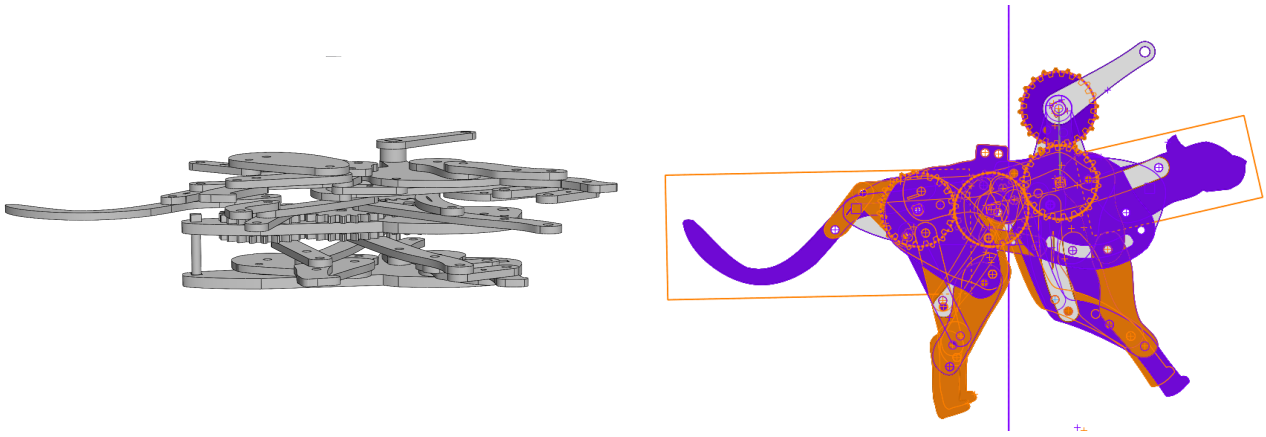
Front-leg geometry refinement and combined leg-kinematics study.

6. Mechanism architecture

The described assembly consists of 24 bars connected through revolute joints and coupled through gears. Two similar linkage sets are mirrored and given a phase difference. Their combination produces coordinated motion while requiring only one independent input.



Side and end views of the complete assembly.



Plan view and geometric construction of the linkage arrangement.

The gear train is important to repeatability: it transfers the input motion and maintains the phase relationship. Joint placement and link lengths determine the path followed by each moving point.

7. Kinematic reasoning and phase coupling

Loop closure

For a planar closed linkage, the oriented link vectors around a loop sum to zero. In component form, the sum of l_i cos(theta_i) equals zero and the sum of l_i sin(theta_i) equals zero. Solving the loop constraints for successive input angles describes the linkage motion.

Mobility

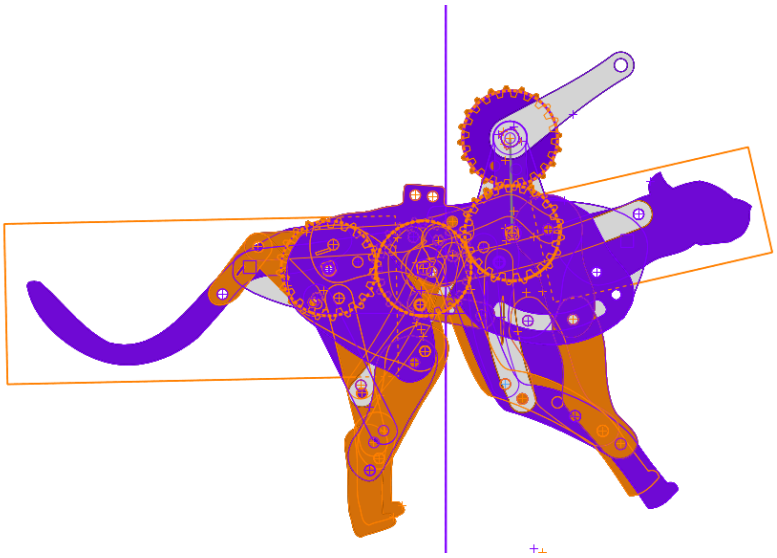
The planar mobility relation M = 3(L − 1) − 2J□ − J□ can be used as a design check when the complete link and joint graph is available. The design uses one independent input; a full mobility analysis requires an inventory of every link, joint and gear constraint.

Coupler trajectories

A tracked point on a rotating link can be expressed as p(theta) = a(theta) + R(phi(theta))r, where a is a joint position, R is the link rotation and r is the point offset in link coordinates. Comparing p over one cycle with the reference path guides geometric refinement.

Phase relationship

The two mirrored sets are mechanically coupled with a phase difference. In an ideal phase model, theta_B = theta_A + delta. The phase offset sets the relative timing of the two linkage sets and must be selected together with the gear ratio and linkage geometry.



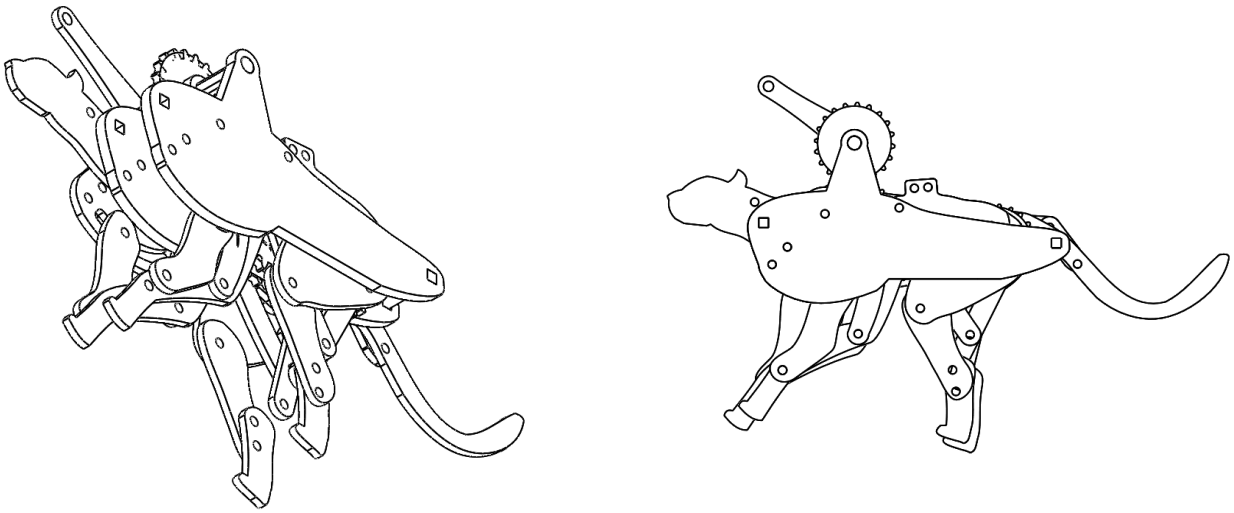
linkage construction used for the motion design.

8. CAD views and assembly development

The CAD views show the side profile, layered plates, joint axes and drive arrangement. They provide a means of checking assembly structure and how the mirrored sets occupy different planes.



assembly viewpoints.



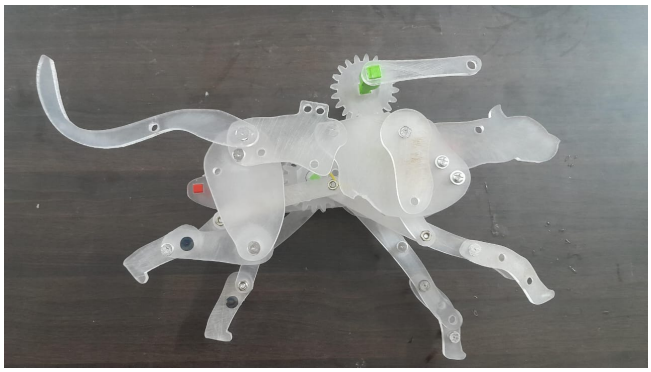
Additional mechanism views.

The complete nine-view set is available on the project page. The model can be requested through the portfolio contact link.

9. CNC-cut prototype

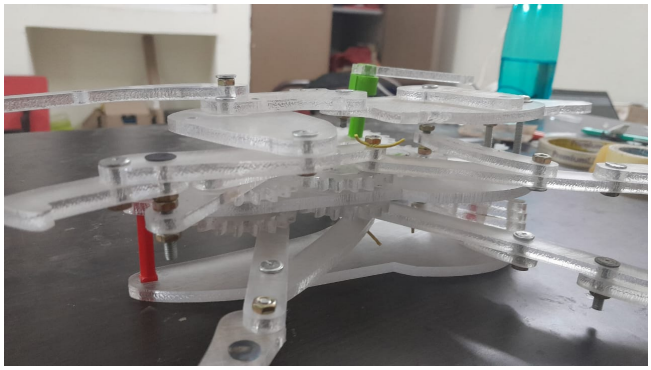
The first physical build uses cut linkage plates assembled with joints and the drive components. The photographs document the transition from a planar design to a layered object that can be moved and inspected.





Cut components and first assembled prototype.





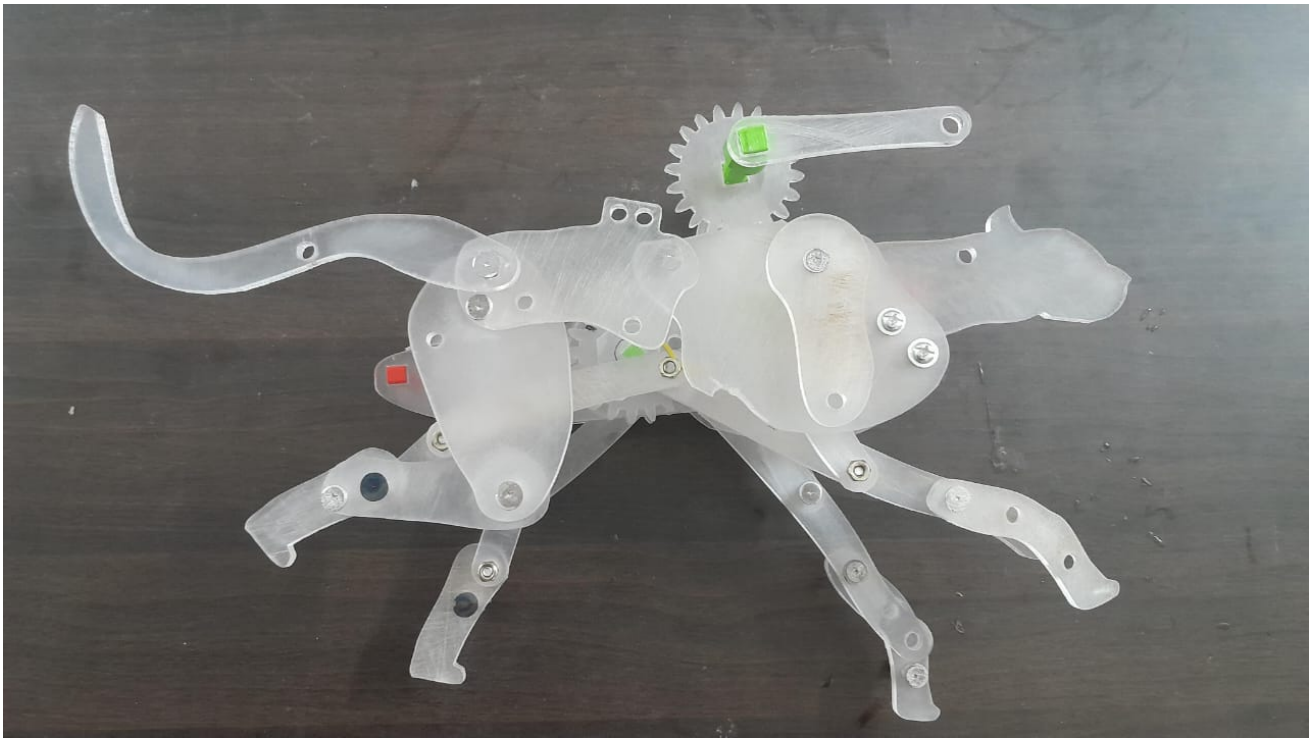
Linkage-chain assembly and additional prototype view.

The working prototype demonstrates the physical mechanism moving. It is useful for observing motion continuity, interference, joint freedom and the visual rhythm of the sculpture.

10. Demonstrations and further development

Recorded demonstrations

The project includes a cheetah gallop reference video, a kinematic motion demonstration and a working-prototype video. The reference footage supports gait observation; the mechanism and prototype demonstrations show the developed mechanical motion.



First physical prototype: the current basis for further mechanical and artistic refinement.

Practical interpretation

The demonstrated result is a kinetic mechanism and a moving prototype. Quantified running speed, ground-contact forces, payload, energy consumption and endurance have not been measured. Speed labels on the animal-reference screenshots refer to the animal reference footage and are not measured speeds of this mechanism.

Next steps

Further refinement can focus on smoother joint motion, reduced interference, repeatable phase setting and the visual balance of the sculpture. A measured trajectory comparison would help distinguish changes that improve path matching from changes made for artistic expression.

11. Project record

Author and designer: Shobit Verma.

Kinetic art

A single-input mechanical sculpture expressing the gathering, extension and recovery of a cheetah gallop.

Gait references

Six reference poses use consistent colours to identify each tracked point. The combined reference brings the complete gait sequence together.

Design development

Front- and rear-leg path studies, geometry refinement, mirrored linkage assemblies and phase coupling connect the observed motion with the mechanical design.

Prototype

CNC-cut plates, articulated joints and drive components form the first physical prototype.

Design filing

Indian design application 519665-001. The filing receipt accompanies this report.